

CSC 411: Machine Learning

Lecture 01: Introduction

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Pan Atlantic University

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What we will cover today

- ❑ Course administration details
- ❑ Why you should learn ML

Course venue

- STT Lab 2

Course prerequisites

- Python Programming
- Basic probability (Bayes Rule, distributions) and Statistics
- Linear Algebra (Vector and Matrix manipulations)
- Calculus

Resources

- <https://github.com/bumie-e/Final-Year-1st-semester-ML-Course/tree/main>

Textbooks for the course

- Python Machine Learning by Sebastian Raschka & Vahid Mirjalili
- Hands-on Machine Learning with Python by Ashwin Pajankar and Aditiya Joshi

Other books for your perusal:

- Pattern Recognition and Machine Learning by Christopher M. Bishop.
- Machine Learning: A Probabilistic Perspective by Kevin P. Murphy.
- The Elements of Statistical Learning by Trevor Hastie, Robert Tibshirani, and Jerome Friedman.
- Machine Learning by Tom Mitchell.

Course Expectations:

- Before each class, the reading materials have been studied. If necessary, code exercises have been done.
- All assignments are submitted at the beginning of the next class.
- Collaboration on assignments is not allowed.
- Labs are designed to be collaborative and engaging and to be done during the class.
- Each completed lab contributes to the participatory mark

Teaching Format

- **Lectures:** Conceptual framing + intuition
- **Labs:** Hands-on notebooks (JupyterLab, VSCode, Google Colab)
- **Case studies:** Real industry scenarios
- **In-class discussions:** Trade-offs, failures, ethics

Grading Structure

- Weekly labs/assignments: 10 marks
- Mid-semester applied assessment: 10 marks
- Final project (team-based): 15 marks
- Participation & engagement: 5 marks
- Exams 65 marks

Resources

- Lecture notes, assignments, reading materials are also available on the page here:
<https://github.com/bumie-e/Final-Year-1st-semester-ML-Course/tree/main>
- Office Hours
 - 1hr/week
 - TBD exactly when

Course Schedules

Weeks	Topic	Assignment
1	What Machine Learning Really Is (and Is Not)	
2	Data: The Real Bottleneck	
3	Features Matter More Than Models	
4	Linear Models and Optimisation	Asst1
5	Non-Parametric Models and Decision Trees	
6	Mid term Assessment	
7	Model Selection, Evaluation, and Learning Theory	

Weeks	Topic	Assignment
8	Support Vector Machines (SVMs) and Kernel Methods	
9	Unsupervised Learning I (Clustering & Dimensionality Reduction)	
10	Reinforcement Learning	Asst2
11	Model Selection and Hyperparameter Tuning	
12	MLOps, Big Data, and Deployment	

Part I: Foundations and Classical Supervised Learning

What Machine Learning Really Is (and Is Not)

What is Machine Learning?

Machine learning is the technology of developing computer algorithms that are able to emulate human intelligence. *El Naqa, I., Murphy, M.J. (2015)*

Part I: Foundations and Classical Supervised Learning

What Machine Learning Really Is (and Is Not)

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A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E . TM Mitchell (1997)

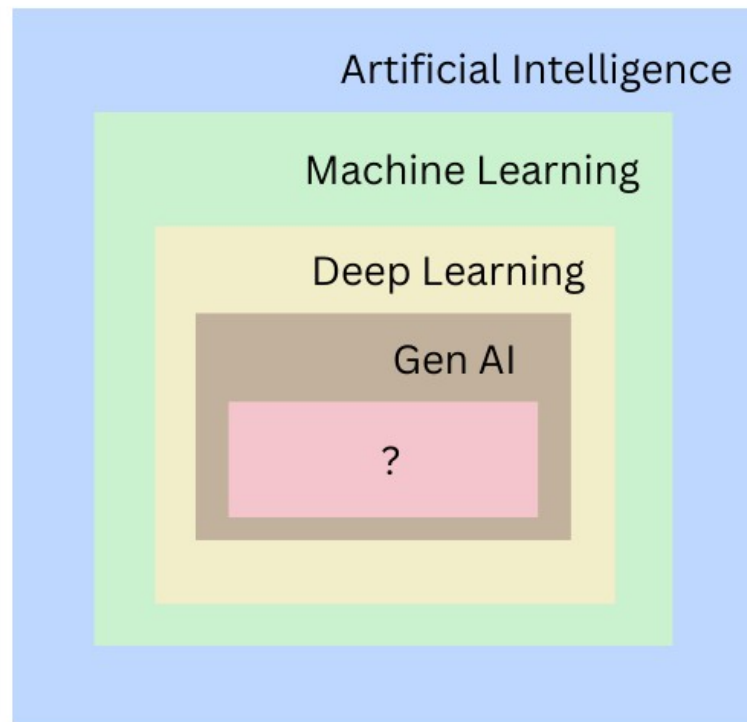
Part I: Foundations and Classical Supervised Learning

What Machine Learning Really Is (and Is Not)

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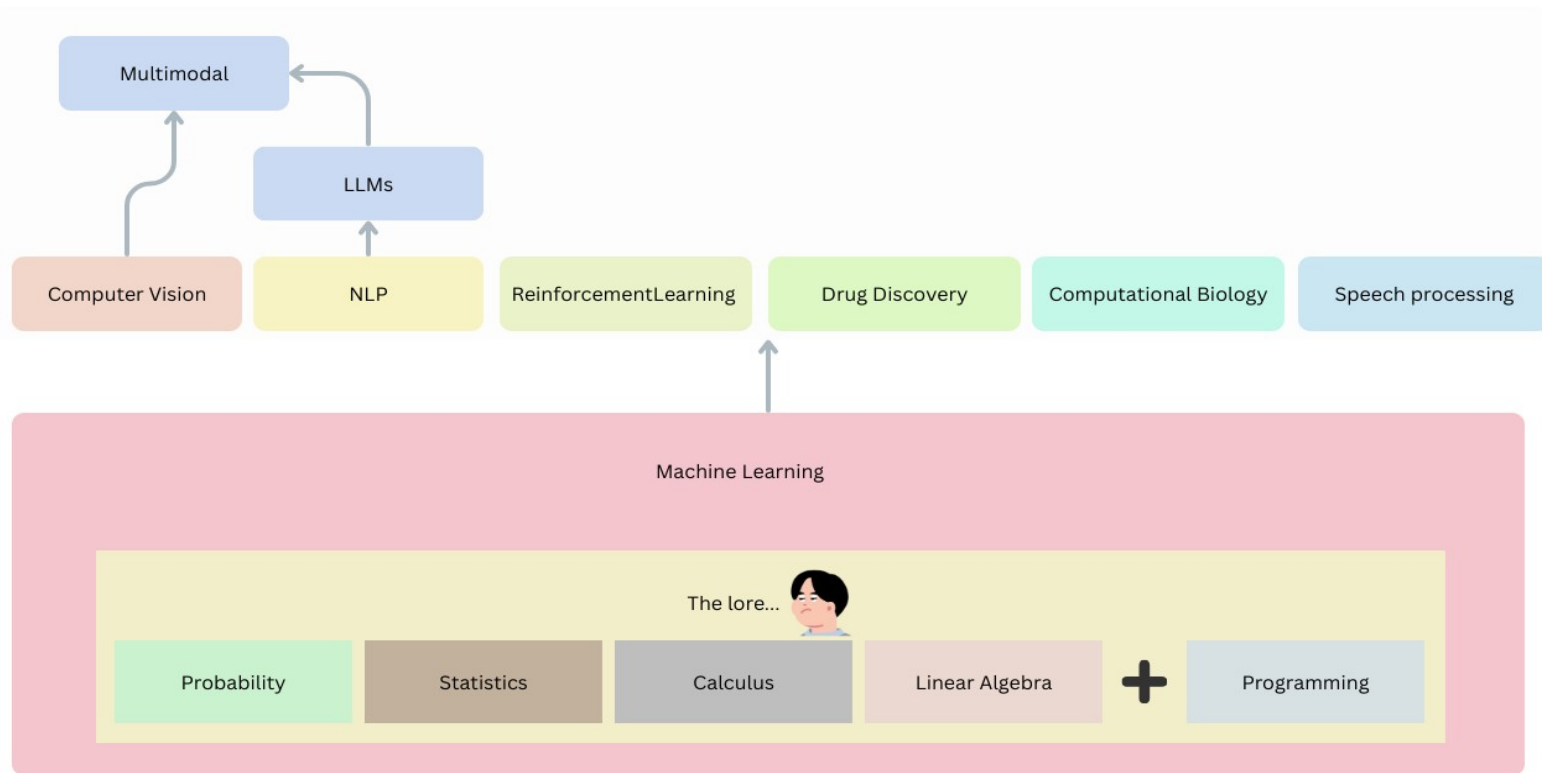
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Part I: Foundations and Classical Supervised Learning

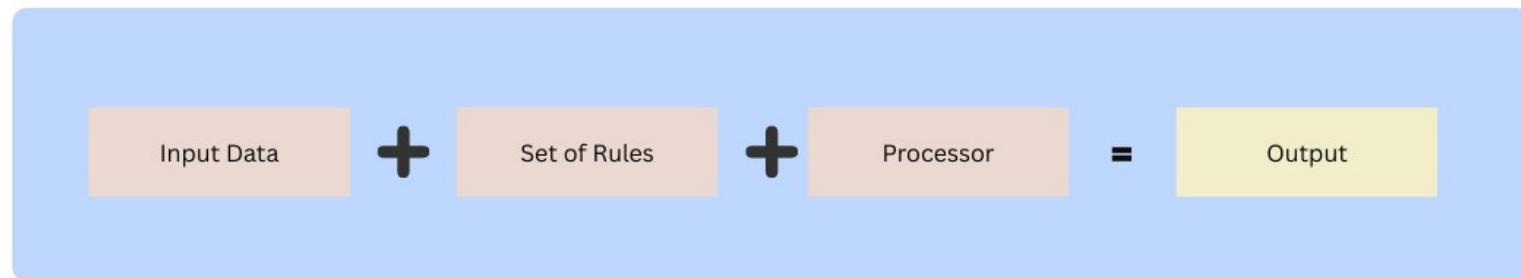
What is Machine Learning?



Part I: Foundations and Classical Supervised Learning

ML vs. Traditional Rule-Based Systems

Typical software development



Part I: Foundations and Classical Supervised Learning

ML vs. Traditional Rule-Based Systems

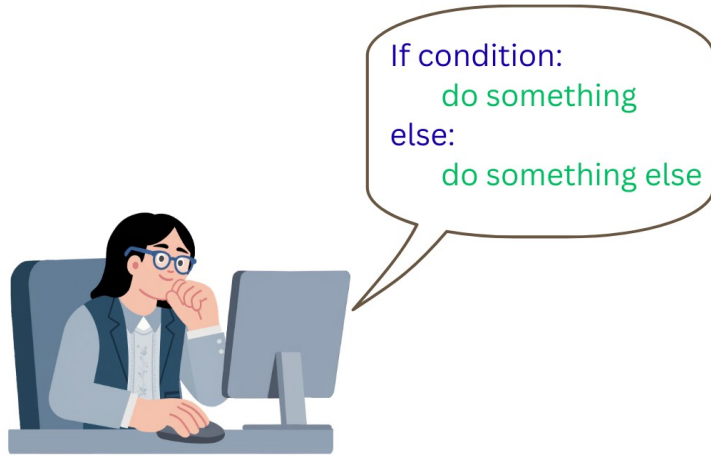
Typical software development



Part I: Foundations and Classical Supervised Learning

ML vs. Traditional Rule-Based Systems

Typical software



Part I: Foundations and Classical Supervised Learning

ML vs. Traditional Rule-Based Systems

Machine Learning

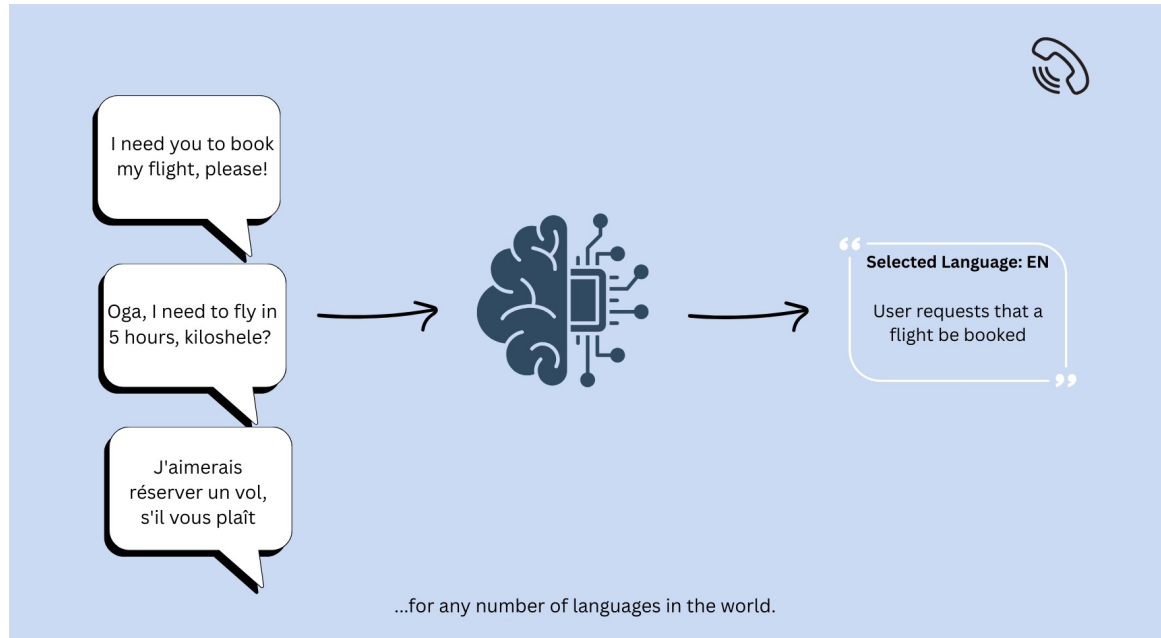


We do not explicitly program the logic; the machine learns the mapping function $f(x)$ to predict y from x

Part I: Foundations and Classical Supervised Learning

ML vs. Traditional Rule-Based Systems

Machine Learning



Part I: Foundations and Classical Supervised Learning

ML vs. Traditional Rule-Based Systems

Why use Machine Learning?

- **Handling Big Data & Complexity:** ML algorithms can analyze massive amounts of data and identify patterns too complex or vast for human analysis.
- **Automation of Tasks:** It automates repetitive, low-cognition tasks, freeing up human resources for more strategic work.
- **Improved Accuracy & Decision Making:** By learning from data, models improve their own performance, leading to more precise predictions.
- **Personalization:** ML powers recommendation engines, allowing companies to offer tailored experiences to users, such as those on Netflix or Amazon.

Part I: Foundations and Classical Supervised Learning

ML vs. Traditional Rule-Based Systems

When to not use Machine Learning?

- **Clear Analytical Solutions:** If a problem can be solved with a simple mathematical formula (e.g., calculating payroll, basic physics simulations) or software techniques.
- **No Data Available:** ML is data-hungry; without relevant experience E , learning cannot occur.
- **100% Accuracy Required:** ML is probabilistic; it makes errors. It is unsuitable for systems where a single error is catastrophic and deterministic rules exist. *Think of an AI Pilot...*

Part I: Foundations and Classical Supervised Learning

Machine learning paradigms

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning
- Semi-Supervised Learning
- Deep Learning
- Active Learning
- Online Learning
- Transfer Learning

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Here, the model learns from a set of inputs and *known* outputs. This input-output set is known as *labelled data*

Goal:

To train a predictive model that learns patterns from this data set, and use it to predict the output of an unseen input

Type	Size	Price
Bungalow	600 by 700	1.2 M
Duplex	800 by 900	4 M



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Can we build a model, that takes inputs x_1 and x_2 , and gives an output/prediction y ?

$$y \approx f(x) = m_1x_1 + m_2x_2 + c$$

Where m is the weight.
 c is the bias

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$$y \approx f(x) = \text{Predict}(x_1, x_2, x_3, \dots, x_n)$$

Given that, **Predict** has already been trained on the set

$$y_1 = x_1 + z_1$$

$$y_2 = x_2 + z_2$$

$$y_3 = x_3 + z_3$$

Can we predict the value of y_4 , given that we have data points $x_3 + z_3$?

Think of:

- House prices, given location and development type (urban vs rural)
- Dog breed, given Picture and behaviour
- Nationality, given Language and Favourite food

Part I: Foundations and Classical Supervised Learning

Machine learning paradigms

What other examples can you think of?

Part I: Foundations and Classical Supervised Learning

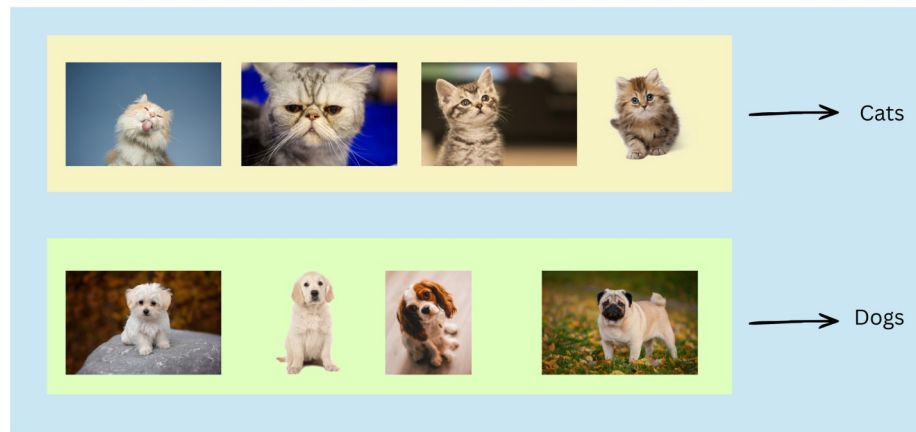
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Classification

Works with discrete categories or classes. These categories are finite.

$$y([0,1,...]) = wx_1 + wx_2 + wx_3 + \dots$$



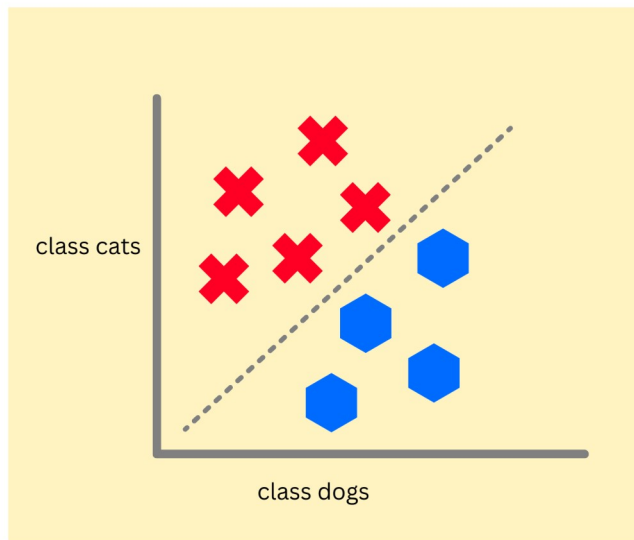
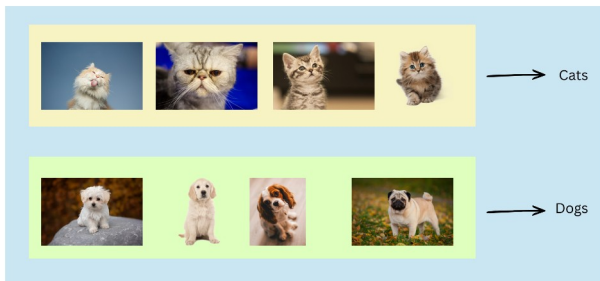
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Classification

- Binary classification, involving two classes
- Multi-class classification, involving more than 2 classes



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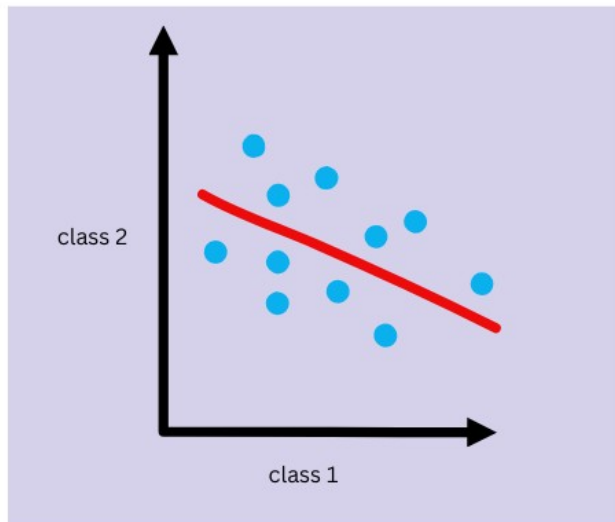
$$y([34.5, 15.2, 33.4]) = wx_1 + wx_2 + wx_3 + \dots + wx_n$$

Where $n \in \mathbb{R}$

Regression

Predicts continuous variables. Cannot be categorised because it'd be infinite.

For example, consider the prices of cars in Lagos



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How does the model know if it's approximating right?

Through a loss function!

A **loss function** calculates how much a model's current subset of outputs deviates from the ground truth while training.

For instance,

Ground truth = 10.3, 2.3, 5.6

Current prediction = 9, 4.1, 3.4

Value of loss = ?

Model Goal:

To minimise loss

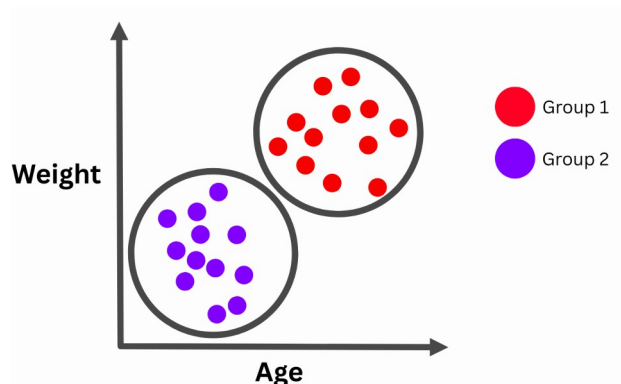
- MAE, MSE, Cross Entropy,

$$\text{Mean Squared Error (MSE)} = \frac{1}{n} \sum_{i=1}^n 2^{-n} (y_i - \hat{y}_i)$$

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Discover underlying structure in a collection of entirely unlabeled data. Only input features x are provided; there are no y labels.

Useful in finding similarities, correlations, or clusters.

Goal:

- To explore patterns, discover hidden features, compress data, or group similar items without explicit labelling

Because unsupervised learning does not acknowledge labels, they don't require loss functions.

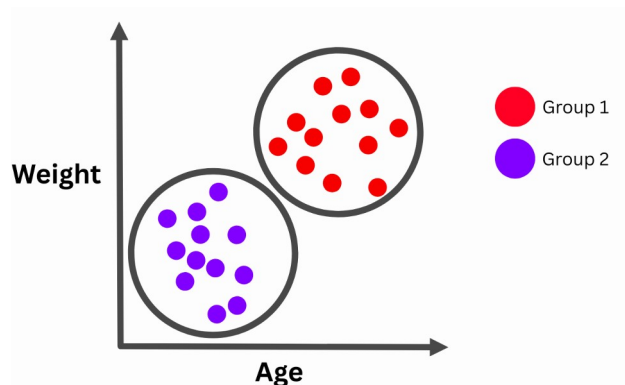
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Common Clustering Algorithms

- Anomaly Detection
- Clustering
- Dimensionality Reduction
- Association Rule Mining



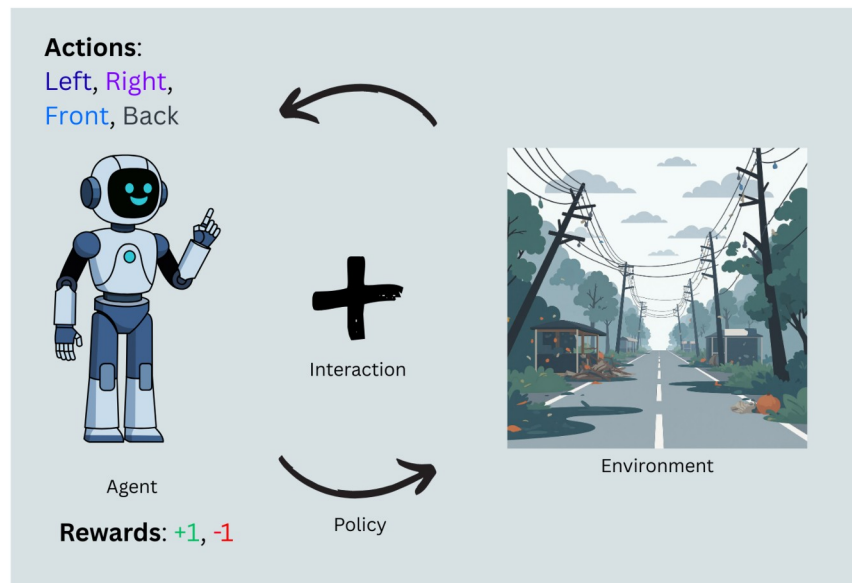
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Teaching agents to learn by trial and error.

Rather than gather a bunch of data to train, data is gotten through interaction with the environment and feedback



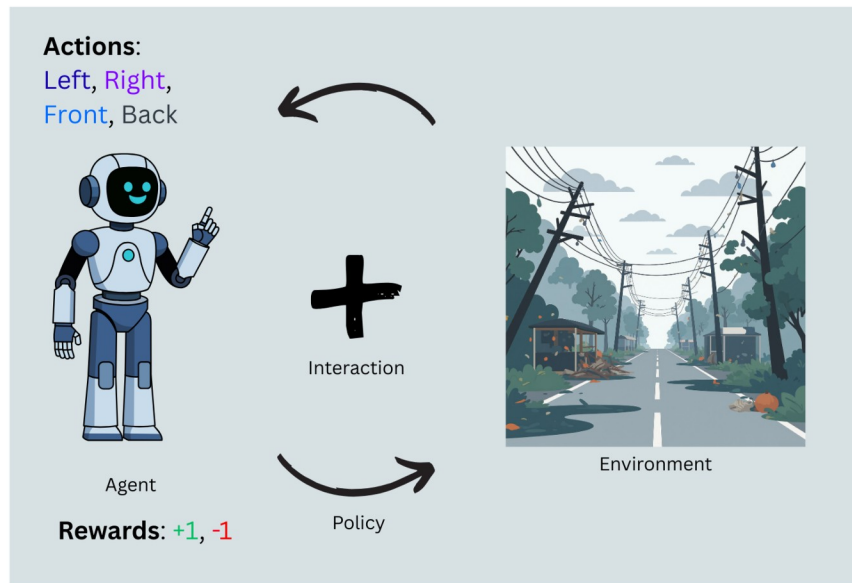
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Can be categorised into:

- Model-based RL
- Model-free RL



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Like supervised learning, some parts of the data contain labels, while the other part doesn't

The goal is still the same, to learn from both. In real world, it is often impractical to label millions of data points

Can be:

- Inductive (learning a general model)
- Transductive (predicting only the specific unlabeled instances)

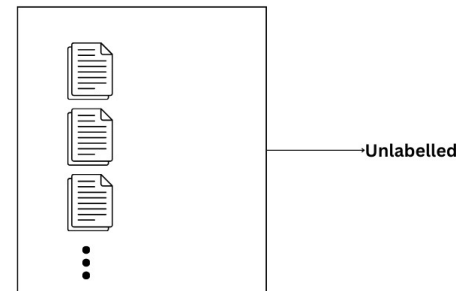
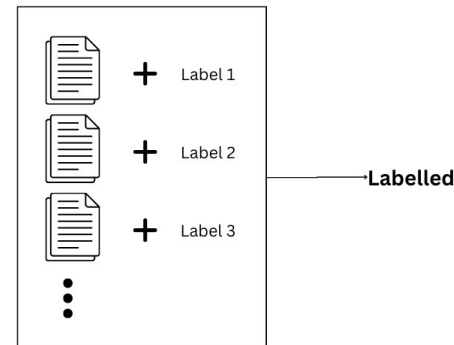
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Can be:

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Like with LLM models today, you'd label a small subset, train a classification model then use it to label the rest

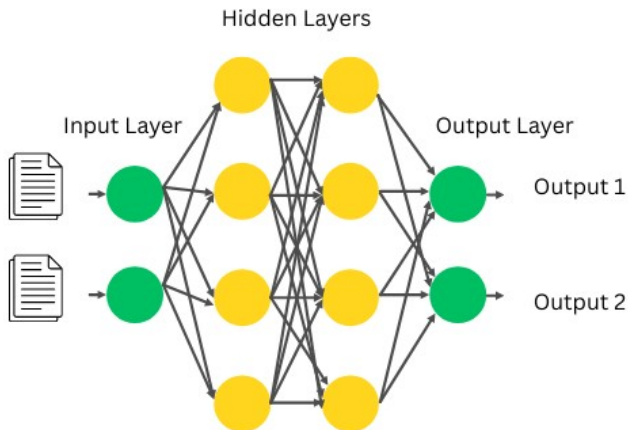
Part I: Foundations and Classical Supervised Learning

Machine learning paradigms

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Inspired by the human brain, the idea of interconnected processing layers, with its own mathematical function (activation), where the output of one layer is input to the other until the final layer.

Through backpropagation, the model can determine the contribution of each node to the loss function by adjusting the weights!



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Machine learning paradigms

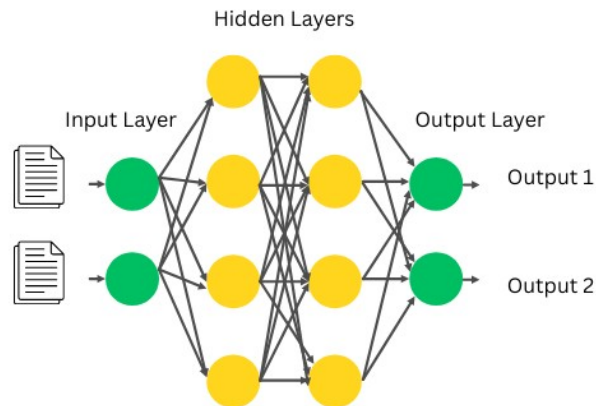
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Activation Function:

A mathematical function that is applied to the output of a neuron.

- Helps introduces non linearity into models.

There are several types, but 3 are most common, Sigmoid, Softmax, and Relu



Sigmoid Function,
$$A = \frac{1}{1 + e^{-x}}$$

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Teaching by learning as you go...

Here,

- the model **selectively chooses the most informative, unlabeled data points to be labeled by a human annotator** (the "oracle") or computer program,
- with the goal of achieving high accuracy with a smaller amount of labeled data,
- thereby saving significant time and cost associated with manual annotation

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How it works:

- Initialise with a "seed" set to train an initial model
- Model predicts on the set of unlabelled data and adds an uncertainty score
- A "query" strategy selects the most valuable data points from the pool for review by the human
- Human expert labels the selected data points
 - This is added back to the existing training set, and retrained.

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Training models using stream of data chunks.

However not, every model can learn online. Models that use a continuous optimisation produce can use online learning For instance, neural networks

*Updates the model based on new information.
Think of stocks, forex trading models...*

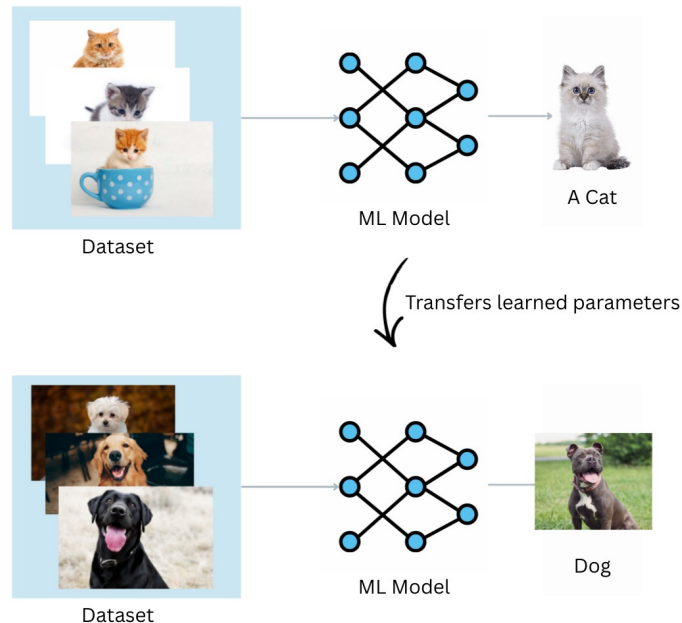
Avoids retraining!

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Involves extracting knowledge from one or more **Source Tasks** and applying it to a different but related **Target Task**.



Why do we need Transfer Learning?

Because the target domain often lacks sufficient training data to train a complex model (like a deep neural network) from scratch.

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Machine learning paradigms

Multi-Task Learning (MTL)

- Training *a single model to solve multiple distinct tasks simultaneously* using shared data representations.
- **Advantage:** Exploits commonalities across tasks, acting as a regularizer to prevent overfitting on any single task

Few-Shot & One-Shot Learning

- Utilizing a large training set to *learn a similarity function*, which is then *used to classify entirely new, unknown classes* using only 1 (One-Shot) or a few (Few-Shot) examples.
- Facial recognition systems, where you provide one photo to your phone, and it learns to recognize you (forever?)

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Industry Applications Today

Computer Vision

➤ Tasks:

- Object detection,
- Semantic segmentation
- Image generation.
- Object tracking
- Boundary Mapping
- Pose Estimation

➤ Used in:

- Autonomous driving
- Facial recognition
- Automated medical scan analysis (e.g., detecting tumors)
- Counting objects

Based on Convolutional Neural Networks (CNNs) which handles spatial pixel structures by sharing weights via filters.

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Industry Applications Today

Natural Language Processing (NLP)

➤ Tasks:

- Machine translation,
- Sentiment analysis,
- Speech-to-text,
- Question answering.
- Text-to-speech
- Translation
- Voice Cloning
- Reasoning
- Text generation

➤ Techniques:

- Word embeddings (Word2Vec)
- N-gram models,
- Recurrent Neural Networks,
- Transformers (LLMs like BERT/GPT).
- LLMs, VLLMs

Used in

- Chatbots, automated document summarization, virtual assistants

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Industry Applications Today

Recommender Systems

➤ Main Idea:

To estimate user affinity for products to drive engagement and sales.

Matrix Factorization (factoring a massive User $\times$ Item matrix to predict missing preferences).

➤ Used Today by:

- Netflix movie recommendations
- Amazon product suggestions
- Spotify playlists

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Why Most ML Projects Fail

Misaligned Business Objectives

Not defining what success looks like or misaligning on KPIs, such as focusing on model accuracy over operational impact.

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Why Most ML Projects Fail

Misaligned Business Objectives

Organizational and Cultural Misalignment

A disconnect often exists between data scientists and business stakeholders

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Why Most ML Projects Fail

Misaligned Business Objectives

Organizational and Cultural Misalignment

Unrealistic Expectations

Overpromising on the capabilities of AI; leads to disappointment when models cannot deliver immediate, perfect results

Part I: Foundations and Classical Supervised Learning

Why Most ML Projects Fail

Misaligned Business Objectives

Organizational and Cultural Misalignment

Unrealistic Expectations

High Costs and Underestimated Resources

When costs (data, infrastructure, labor) outweigh the projected ROI, projects are often terminated.

Part I: Foundations and Classical Supervised Learning

Why Most ML Projects Fail

Misaligned Business Objectives

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Unrealistic Expectations

High Costs and Underestimated Resources

Lab vs Real World!

Many models work well in a lab environment but fail to **woefully** in production

Part I: Foundations and Classical Supervised Learning

Overview of the Course Project

Project Overview & Objective

- Frame a vague or complex real-world challenge as a formal machine learning task, build a reproducible ML pipeline, and rigorously evaluate your solution while acknowledging ethical and practical limitations.

The problem of choice should relate to a personal experience, a challenge within your local community, or an environmental issue.

Part I: Foundations and Classical Supervised Learning

Overview of the Course Project

Project Overview & Objective

- Frame a vague or complex real-world challenge as a formal machine learning task, build a reproducible ML pipeline, and rigorously evaluate your solution while acknowledging ethical and practical limitations.

Team Size:

- You will work in small groups of 5 students.

Scope:

Your project must integrate multiple facets of practical machine learning, including

- data preprocessing,
- baseline establishment,
- algorithm implementation,
- regularization,
- model selection,
- error analysis

Part I: Foundations and Classical Supervised Learning

Lab

Setup coding Environment

- Google Colab
- Anaconda environments

Thank you.

Questions?